

Multimodal Learning for Wearable Health Data: From Raw Signals to Clinical Insight

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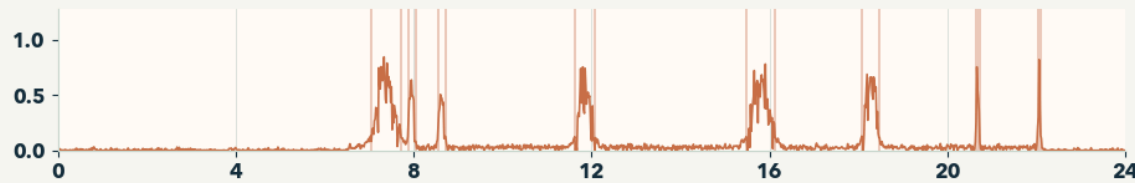


From wearable device data to clinical insight

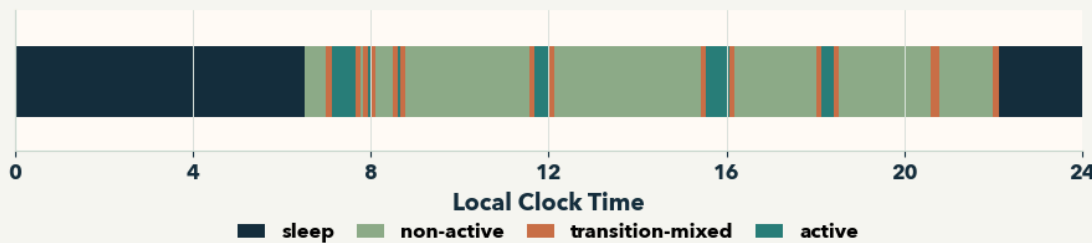
Synthetic 24-hour wearable sample data

Signals, labels, and model probabilities are simulated for a RISE-like participant; no real participant data shown.

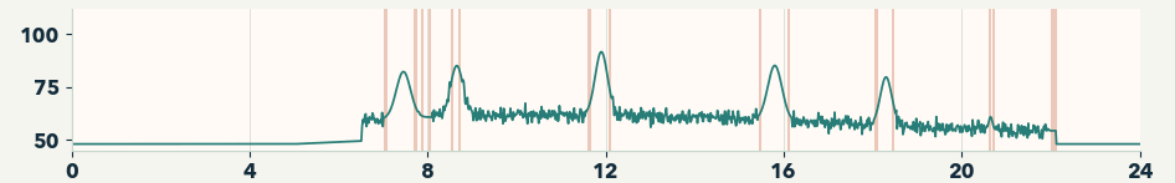
Raw Sensor Stream



Derived Posture / Transition Labels



Heart-Rate Context



Model-Inferred State Probabilities



The goal is to turn noisy longitudinal streams into phenotypes that can support monitoring, prediction, and intervention evaluation.

Challenges in wearable AI

Free-living measurement

Measurements are often recorded during daily life, without laboratory calibration or controlled task labels.

Proxy or absent labels

Gold-standard labels are often unavailable. Reference labels may come from another device, an algorithm, or limited manual annotation.

Missing data

Sensor glitches, battery depletion, participant non-compliance, wear time, and device placement all change what is observed.

Clinical sample constraints

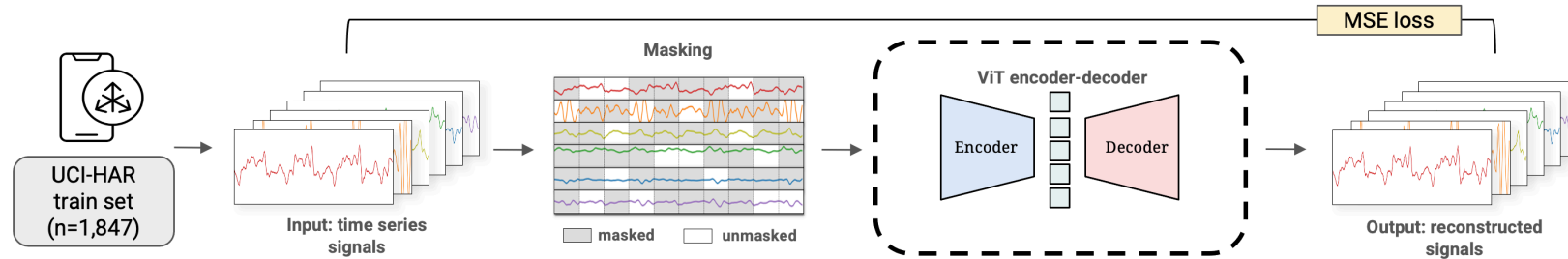
Calibrated studies are expensive, so small sample sizes and sparse health outcomes are common.

Heterogeneity

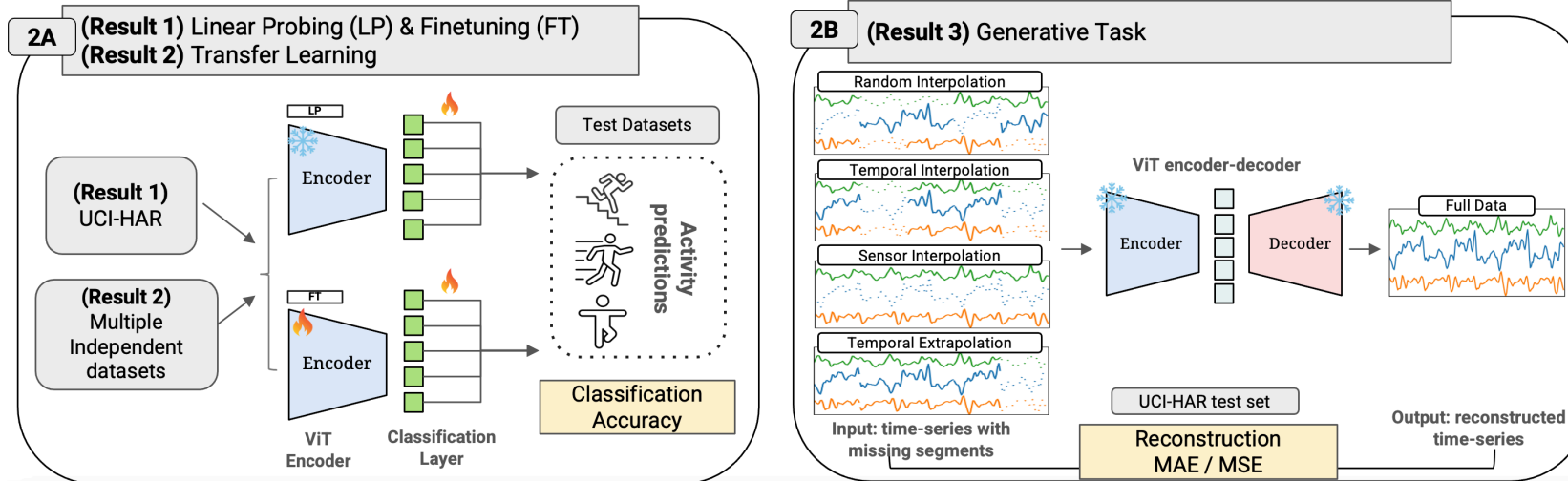
Devices, body sites, populations, comorbidities, and study protocols can all shift the data distribution.

MoCA: Multi-Modal Cross-Masked Autoencoder

1. Pre-training (self-supervised learning: reconstruction)



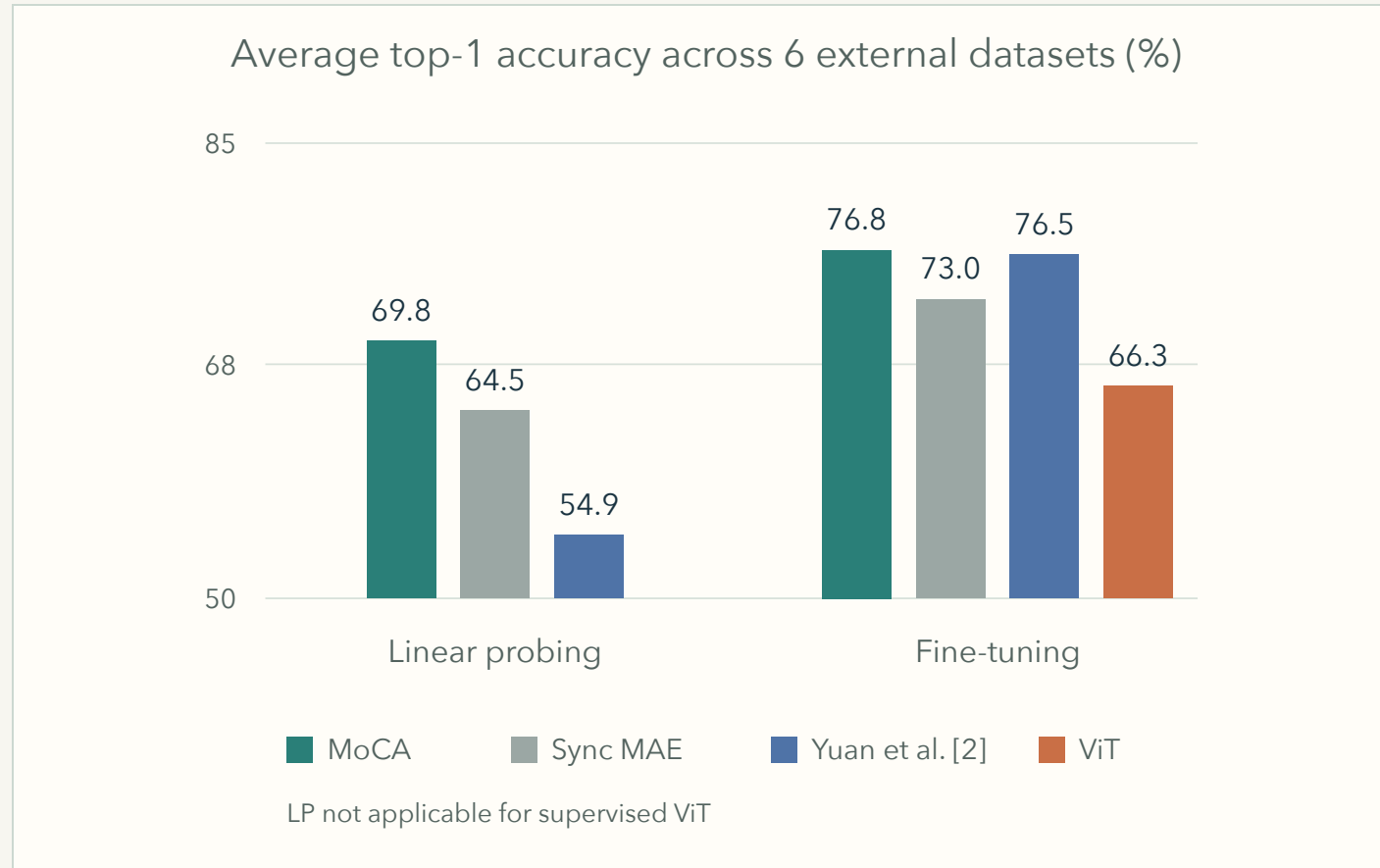
2. Downstream Tasks



MoCA [1] uses principled cross-modality masking within a masked autoencoding SSL framework to leverage correlation structure across sensor modalities.

The method is theoretically grounded by connecting multimodal MAE loss to kernelized canonical correlation analysis through an RKHS framework, providing guidance for correlation-aware masking design.

MoCA transfers across sensor studies



Key results

- On UCI-HAR, MoCA reaches **96.6%** sample-wise fine-tuning accuracy and **91.36%** subject-wise 5-fold cross-validation accuracy.
- Across six external datasets, frozen MoCA representations achieve **69.8%** average linear-probing accuracy, above Sync MAE and Yuan et al. [2].
- After fine-tuning, MoCA reaches **76.8%** average accuracy, above Sync MAE and supervised ViT, and comparable to Yuan et al. [2].
- The frozen-encoder result supports MoCA as a reusable wearable representation learner, not only a task-specific classifier.

MoCA manuscript [1]: UCI-HAR benchmark and transfer learning results across WISDM, IMWSHA, OPPORTUNITY, ADL, PAMAP2, and RealWorld.

RISE Study: Design and Goal

Study design

The Rise for Health Study [3] is a three-arm, three-month randomized trial involving **407** postmenopausal women, with mean age **67.81**.

Wearable data from two concurrent devices

The dataset contains concurrent thigh-worn **activPAL** and hip-worn **ActiGraph GT3X+** tri-axial accelerometer readings.

Learning Intervention Targeted Behaviors

The trial's interventions **targeted less sitting** and **more sit-to-stand transitions**. A binary sitting classifier can miss the mechanism the intervention is trying to change.

Analysis goal

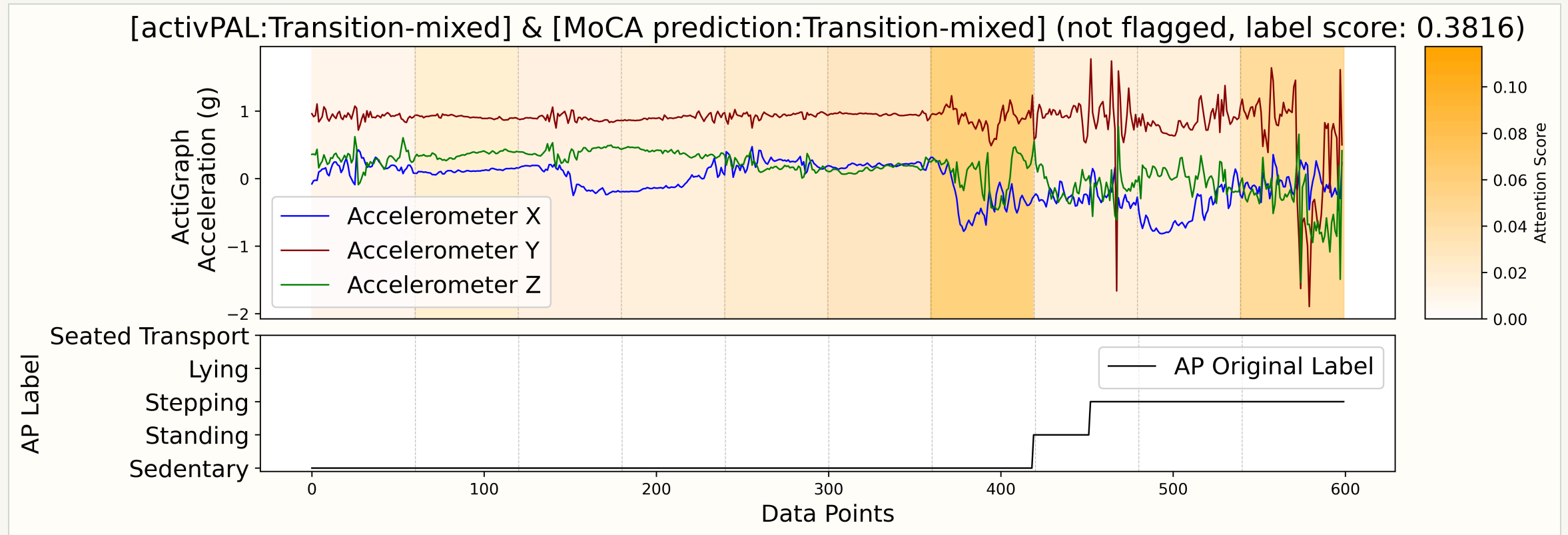
Use MoCA to learn transition-aware activity labels from hip-worn ActiGraph signals, then evaluate whether model-derived labels preserve intervention effects seen with activPAL reference labels.

RISE inference pipeline



The practical goal is to test whether MoCA-derived labels can preserve the intervention signal while reducing reliance on thigh-worn activPAL labels as the ground-truth reference.

MoCA in RISE focuses on the seconds when posture changes



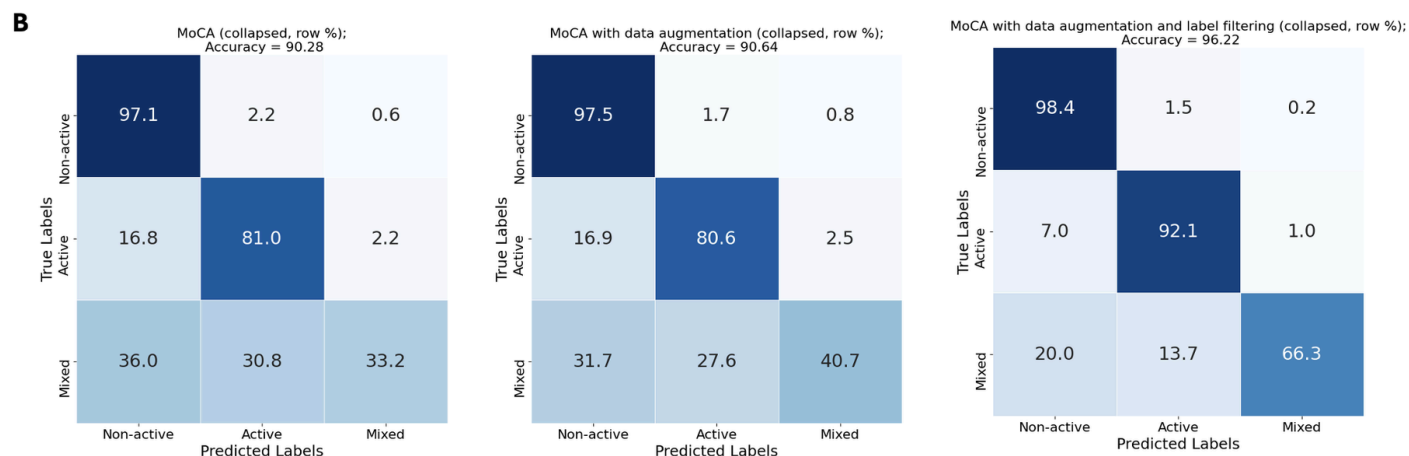
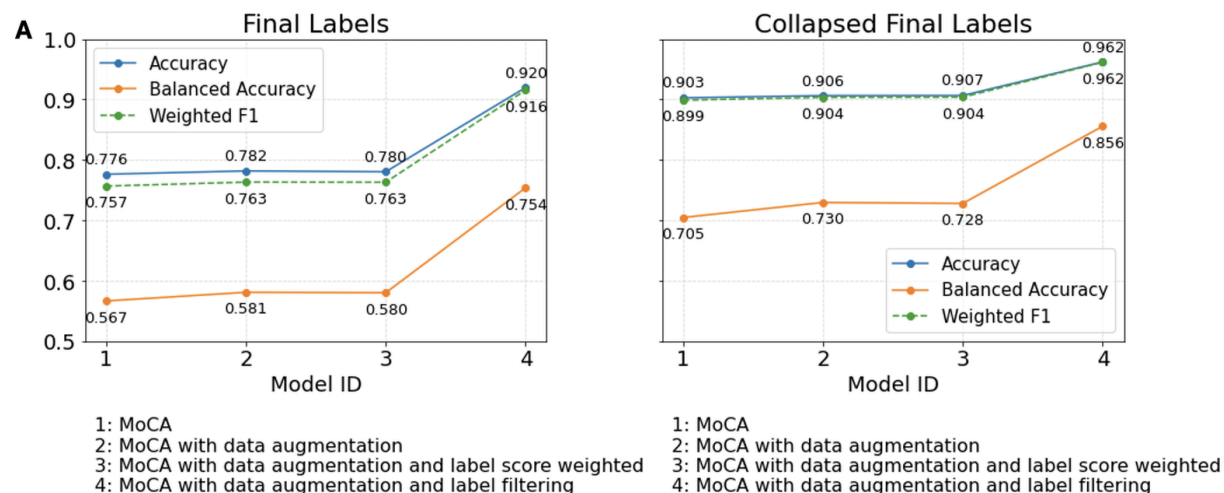
Model focus

The attention map highlights the short parts of the 20-second window where the ActiGraph signal shifts, the likely sit-to-stand or posture-change moments.

Why transitions are hard

A transition window can contain more than one behavior: sitting, standing up, and movement. That makes the label less clear than a pure sitting or pure active window.

Label filtering improves transition recognition



Results

92.0% seven-label accuracy and **96.2%** collapsed-label accuracy.

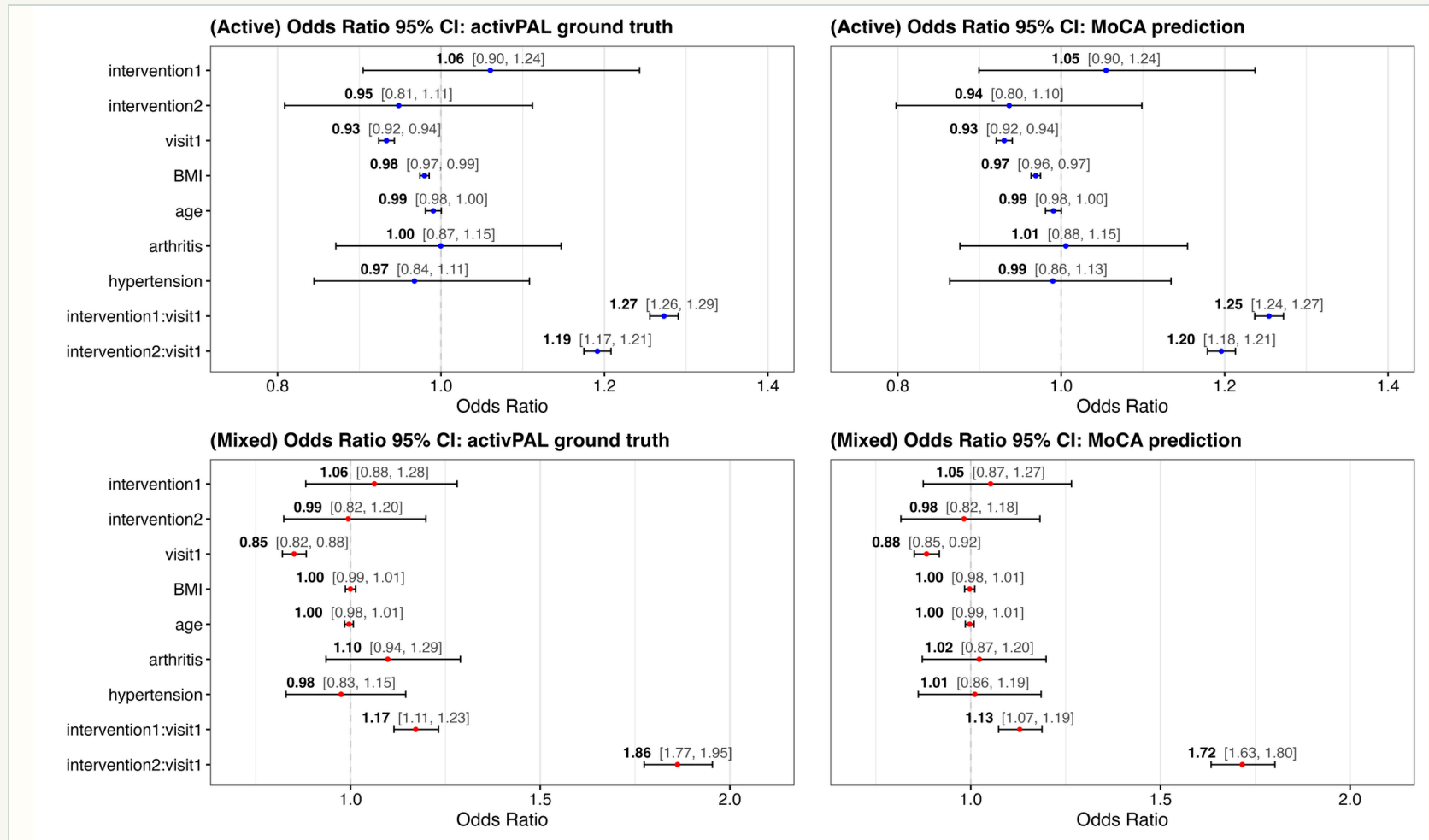
Mixed transitions

Sensitivity for the Mixed label nearly doubles, from **33.2%** to **66.3%**, after data augmentation plus label filtering.

Interpretation

Cleanlab-derived confident learning [4] targets interval-level proxy-label uncertainty before training the deep model.

Model-derived labels reproduce intervention effects



Less-sitting arm

Active behavior shows similar direction and magnitude under activPAL and MoCA labels.

Sit-to-stand arm

Mixed-transition behavior shows the largest increase in the arm designed to increase sit-to-stand transitions.

Classification matters only if it preserves the study contrast that the trial was designed to estimate.

Remaining challenges

Proxy labels

Reference-device labels are not directly observed truth. Around posture changes, a 20-second window can mix sitting, standing, and movement, so training pipelines should flag labels with low confidence and clean them before training.

Missingness and adherence

Wear time, compliance, body placement, and battery life can carry health and behavior information.

Longitudinal leakage

Participant-, day-, site-, and intervention-aware splits are required for honest generalization claims.

Clinical estimands

Model selection should be tied to health outcomes and intervention contrasts, not only interval-level accuracy.

Field direction: world models, representation learning, endpoint definition, validation design, and deployment constraints need to be planned together.

Message for Future Studies

- 1 Go beyond classification of short windows toward longitudinal phenotypes, trajectories, and clinically interpretable endpoints.
- 2 Pair model outputs with careful inference, using proper statistical design and data analytics for the study question.
- 3 Use multimodal data for insight, with careful preprocessing, fusion, and disentanglement of modality-specific and shared signals.
- 4 Build close collaborations with clinicians, domain researchers, and guideline makers so model targets match decision needs.

Acknowledgments

Research team members

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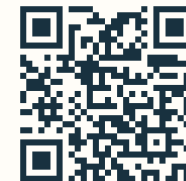
Clinical Study Collaborations

RISE for Health
SOL-VIDA
OPACH / OPACH2
WHISH
iWATCH
I Can!
study teams and participants

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Thank you

Questions, discussion, and collaborations welcome



MoCA manuscript
arXiv:2506.02260

References

- [1]** Ryu H, Chen Y, Wang Y, LaCroix AZ, Di C, Natarajan L, Wang Y, Zou J. *Multi-modal Cross-masked Autoencoder for Time Series in Digital Health*. arXiv:2506.02260, 2025.
- [2]** Yuan H, Chan S, Creagh AP, Tong C, Acquah A, Clifton DA, Doherty A. Self-supervised learning for human activity recognition using 700,000 person-days of wearable data. *NPJ Digital Medicine*. 2024;7(1):91.
- [3]** Hartman SJ, Dillon LW, LaCroix AZ, Natarajan L, Sears DD, Owen N, Dunstan DW, Sallis JF, Schenk S, Allison M, et al. Interrupting sitting time in postmenopausal women: protocol for the Rise for Health randomized controlled trial. *JMIR Research Protocols*. 2021;10(5):e28684.
- [4]** Northcutt C, Jiang L, Chuang I. Confident learning: Estimating uncertainty in dataset labels. *Journal of Artificial Intelligence Research*. 2021;70:1373-1411.